

Meeting 4

This is the summary of the first meeting held with representative from institute C. The table summarizes the purposes, main topics discussed, and outputs of the meetings.

Fourth meeting	
Stakeholders	Institute C; representative C1 and C2.
Topics of discussion	We initiated discussions regarding defining the domain of reference of the ClimOO Ontology. We engaged with two stakeholders regarding perspectives on sustainability and related frameworks of reference, agri-food systems of interest and related boundaries, primary features to address with regard to plastic, plastic pollution, and climate change, and possible data to interoperate.
Stakeholder perspective on sustainability	Sustainability in terms of the Environmental, Social, and Governance (ESG) principles. Focus on Environment and Society.
Additional take-home messages regarding sustainability	- Plastic pollution and climate change as antagonists to achieving sustainability. - Sustainable production: production that ensures resource recovery and benefits beyond solely human benefits, e.g. benefits for soil. With regard to agriculture: practices of bio-agriculture, e.g. food production without pesticides.
Framework of reference on sustainability, plastics, and climate change	XXXX
Relevant system stakeholders	To further discuss with other stakeholder institutes.
Agriculture system of reference	To further discuss with other stakeholder institutes.
Primary features to address concerning plastic pollution and climate change	- To date, there has been an overuse of traditional plastics. - No material can currently substitute plastics adequately. - There is a need to implement alternative solutions, e.g. fostering recycling. - Plastics and microplastics are everywhere, including oceans and water streams, soils, and human bodies.

	• Neglect of the consequences of the usage of plastics due to their large versatility and usefulness across sectors and domains. • Currently, bio-based and biodegradable plastics are seen as alternatives to traditional plastics, but the scenario may change due to the ongoing debates, especially within the EU, on their contribution to global plastic pollution. ○ Presence of lobbies that antagonize this process. ○ Their production process is in more demand than that of traditional plastics. ○ There is a need for more adequate regulations and definitions. • Attention to climate change, especially with regard to agriculture. • Emphasis on the interrelation between climate change and plastic pollution. ○ For example, the usage of plastic coverage for cultivation climate protection leads to the release of microplastics into the soil. • Need to cover implications of climate change for humans and beyond, e.g. for animals and the environment. • To account for both direct and indirect effects of climate change in agriculture. • Currently, to have a general emphasis on sustainability goals, assessment methods, and indicators by now, e.g. on SDGs. No strict need to represent them in the ontology by now. Now need to focus on the circular economy by now. In the ontology: • To pay attention to the different systems of classification of plastics and microplastics. • To cover different types of plastic and plastic debris, e.g the account based on particle size. ○ Emphasis on micro- and nanoplastic particles. • To include references to plastic additives and plastic and plastic debris sorption of pollutants. A general focus is adequate by now. • Attention to the distinction between bio-based and biodegradable plastics and related debates on current standards.
Data of reference	To further discuss later when the project is more advanced.
Relevant projects	XXXX